

AI/ML Full Stack Developer

Summary

GenAI Engineer with **5 years of experience** in developing and deploying **AI/ML and Generative AI solutions**. Proficient in **Python, PyTorch, TensorFlow, LangChain, and Hugging Face**. Skilled in **JavaScript/TypeScript, React, Node.js**, and cloud platforms (**AWS, Azure, GCP**) to deliver scalable AI-driven solutions, **NLP, and end-to-end data pipelines**. Experienced in delivering **scalable AI systems**, integrating ML models with full-stack applications, and collaborating across teams to solve complex business problems.

Technical Skills

Programming Languages: Python, React, Javascript SQL, Java, C++, Scala

Generative AI & Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers, LangChain, OpenAI API, Matplotlib, Seaborn

Natural Language Processing (NLP): NLTK, spaCy, Transformers, LangChain, RAG (Retrieval-Augmented Generation) pipelines, embeddings

Big Data & Distributed Computing: Apache Spark, Hadoop, Kafka, HBase, PySpark

Cloud Platforms & AI Services: AWS (SageMaker, Lambda, S3), Microsoft Azure (Azure OpenAI, Cognitive Services, ML Studio), Google Cloud Platform (Vertex AI, AI APIs)

Data Pipelines & ETL Tools: Apache Airflow, Kafka, Informatica, Palantir, PySpark

Data Visualization & Reporting: Tableau, Power BI, Plotly, Dash, Excel

Database Systems: MySQL, PostgreSQL, NoSQL (MongoDB, Cassandra)

DevOps & CI/CD: Docker, Kubernetes, Jenkins, Azure DevOps, GitHub Actions

Specializations: Generative AI, Large Language Models (LLMs), Machine Learning, Deep Learning, Artificial Intelligence, NLP, Data Science & Analytics, Software Engineering

Collaboration & Soft Skills: Cross-functional teamwork, problem-solving, strong communication, and project leadership

Education

University of Texas Arlington, United States

Aug 2023 - May 2025

Master of Science, Data Science

Work Experience

Verizon

May 2024 – Present

Gen AI Full Stack Developer

United States

- Designed, developed, and deployed **Generative AI applications** on **Azure OpenAI Service**, leveraging GPT models for text generation, summarization, and semantic search.
- Built **enterprise AI solutions** using **Azure Cognitive Services** (Text Analytics, Translator, Speech-to-Text, Custom Vision) and **Azure Machine Learning** for model training and lifecycle management.
- Integrated **LangChain, Semantic Kernel, and RAG pipelines** with **Azure Cognitive Search** and **Azure Storage** for document retrieval and context-aware responses.
- Created **custom copilots** and **autonomous AI agents** using **Microsoft Copilot Studio, Power Virtual Agents, and Azure Bot Service**, enabling business workflow automation.
- Developed **vector embeddings** using **Azure OpenAI and FAISS/Weaviate** for semantic search and question-answering use cases.
- Managed **end-to-end MLOps pipelines** on **Azure ML**, incorporating **MLflow, Databricks, and Delta Lake** for continuous integration and delivery.
- Implemented **fine-tuning of LLMs** using domain-specific data with **Azure OpenAI fine-tuning APIs**, ensuring accuracy and alignment with enterprise needs.
- Automated **document ingestion, data labeling, and preprocessing workflows** using **Azure Data Factory, Event Hubs, and Azure Functions**.
- Secured AI workloads using **Azure Key Vault, RBAC, VNet integration, and Managed Identities** for enterprise compliance.
- Monitored application performance and **optimized cost** using **Azure Monitor, Application Insights, and Cost Management tools**.
- Built **scalable APIs and microservices** for AI-driven solutions using **Azure Kubernetes Service (AKS)** and **Azure API Management**.
- Collaborated with cross-functional teams to **translate business requirements into AI-driven workflows**, ensuring production-

grade reliability.

- Worked with **Python (FastAPI, Flask)**, **JavaScript/TypeScript**, and **Power Platform tools** for application development and integration.
- Leveraged **Azure DevOps pipelines** for CI/CD automation and streamlined deployment cycles.
- Built **responsive full-stack dashboards** using **React**, **TypeScript**, **Node.js**, and **cloud services**

Cigna
2024

AI/ML Engineer

September 2023 – March

Texas, United States

- Developed and deployed **predictive analytics models** on **Azure Machine Learning** to forecast patient readmission risk, improving care outcomes and reducing costs.
- Built **Generative AI pipelines** using **Azure OpenAI Service**, **LangChain**, and **Azure Cognitive Search** to summarize clinical notes and automate prior authorization workflows.
- Implemented **HIPAA-compliant data pipelines** using **Azure Data Factory**, **Event Hubs**, and **Databricks** for large-scale ingestion of claims and EHR data.
- Fine-tuned **GPT-based LLMs** with domain-specific healthcare datasets to enhance accuracy in clinical document processing.
- Deployed **microservices on Azure Kubernetes Service (AKS)** with **CI/CD pipelines** using **Azure DevOps** for production reliability.
- Used **MLflow** and **Delta Lake** for versioning, model monitoring, and reproducible experiments.
- Integrated **Azure Cognitive Services (Text Analytics, Translator)** to extract medical terms and classify unstructured data.
- Collaborated with compliance teams to implement **enterprise-grade security controls** (Key Vault, RBAC, VNet integration).
- Designed dashboards with **Power BI** to visualize model predictions, improving decision-making for care management teams.

HCL Tech
2023

Data Scientist

June 2021 – May

India

- Built **scalable ETL pipelines** using **AWS Glue**, **Lambda**, and **Step Functions** to process high-volume healthcare claims and EHR data.
- Developed and deployed **machine learning models on Amazon SageMaker** (XGBoost, PyTorch) to predict patient readmission risk and improve care management workflows.
- Implemented **transformer-based NLP models** using **Hugging Face on SageMaker** to extract clinical entities and summarize unstructured medical text.
- Delivered **HIPAA-compliant APIs** with **API Gateway**, **Lambda**, and **DynamoDB**, enabling secure integration of ML predictions into enterprise systems.
- Created **full-stack dashboards** using **React**, **TypeScript**, and **AWS Amplify** to provide clinicians and analysts with real-time model insights.
- Automated **CI/CD pipelines** using **AWS CodePipeline**, **CodeBuild**, and **CodeDeploy** to streamline deployment of AI and data services.
- Applied **IAM roles**, **VPC**, and **KMS encryption** to ensure compliance and secure patient data across all workloads.
- Monitored production workloads using **CloudWatch** and **X-Ray** to maintain high availability and optimize resource usage.
- Collaborated with product, data, and engineering teams to deliver **end-to-end AI/ML solutions** from data ingestion to user-facing applications.

Projects

AI Chatbot for Student Queries

- Developed an **NLP-based chatbot** using **Rasa** and **Python** to answer university-related questions in real time.
- Integrated with a database to provide **personalized responses** and **track frequently asked queries**.

Voice Activated Virtual Assistant

- Built a speech-to-text and text-to-speech assistant using **Google Speech API** and **NLP techniques**.
- Implemented task automation features like **reminders**, **weather updates**, and **web search**.